arXiv 2305.10601
Venue: NeurIPS 2023

Tree of Thoughts: *Deliberate Problem Solving with Large Language Models*

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Background

Standard LLMs are...

- autoregressive, left to right token predictors
- **BAD** at backtracking, exploration, fixing errors

Idea: Branching exploration **(tree)** improves final result

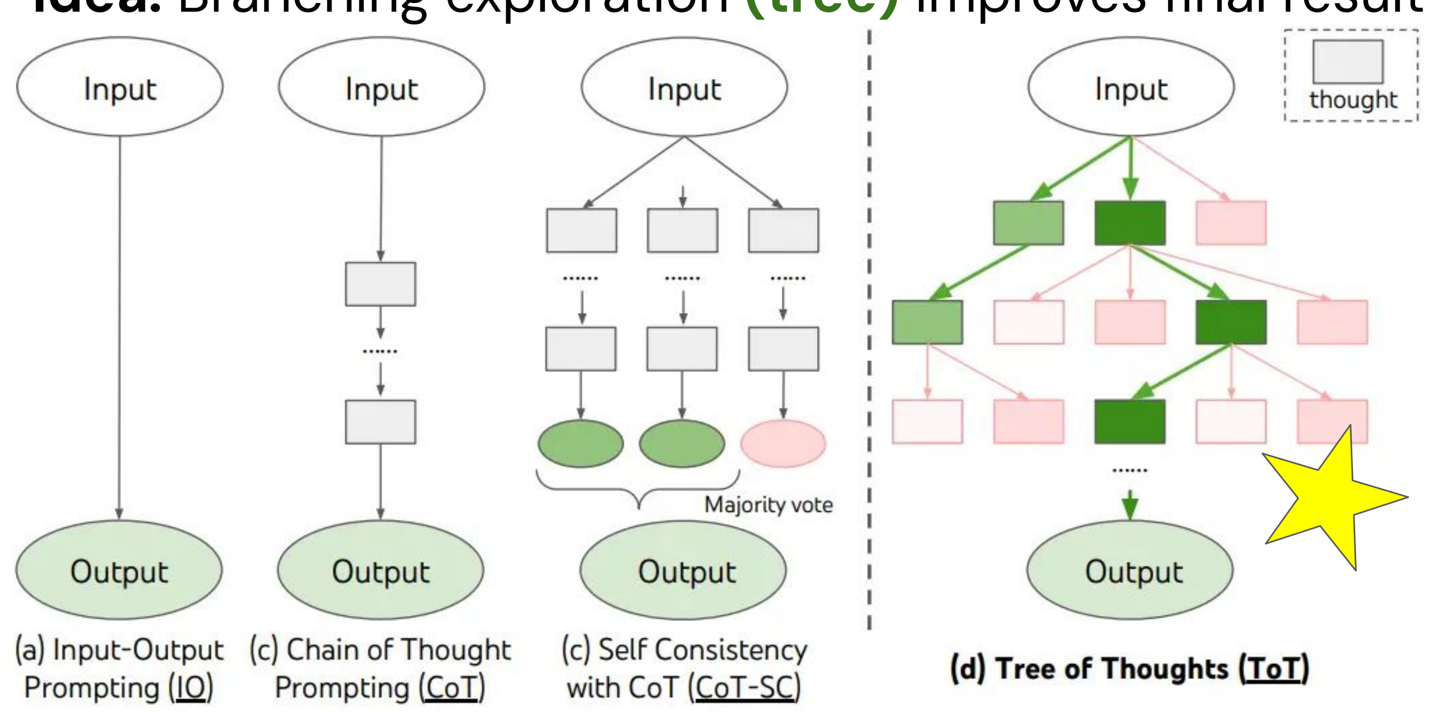


Fig 1. Yao et al. (2023) Prompting techniques using reasoning.

Paper Findings

Method	Success
IO prompt	7.3%
CoT prompt	4.0%
CoT-SC (k=100)	9.0%
ToT (ours) (b=1)	45%
ToT (ours) (b=5)	**74%**
IO + Refine (k=10)	27%
IO (best of 100)	33%
CoT (best of 100)	49%

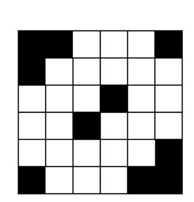
Table 2: Game of 24 Results.

4nums.com

- 100 difficult games
- mathematical reasoning

randomwordgenerator.com

- 100 sentences
- creativity



goobix.com

- 156 5x5 boards
- problem solving

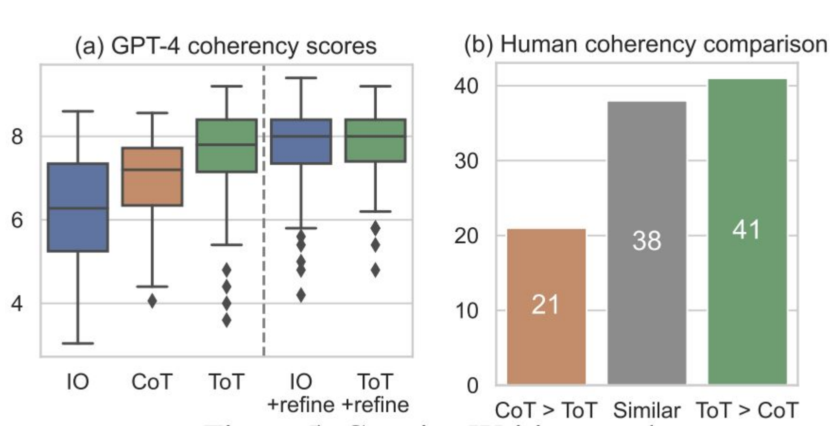


Figure 5: Creative Writing results.

Method	Success Rate (%) Letter	Word	Game
IO	38.7	14	0
CoT	40.6	15.6	1
ToT (ours)	**78**	**60**	**20**
+best state	82.4	67.5	35
-prune	65.4	41.5	5
-backtrack	54.6	20	5

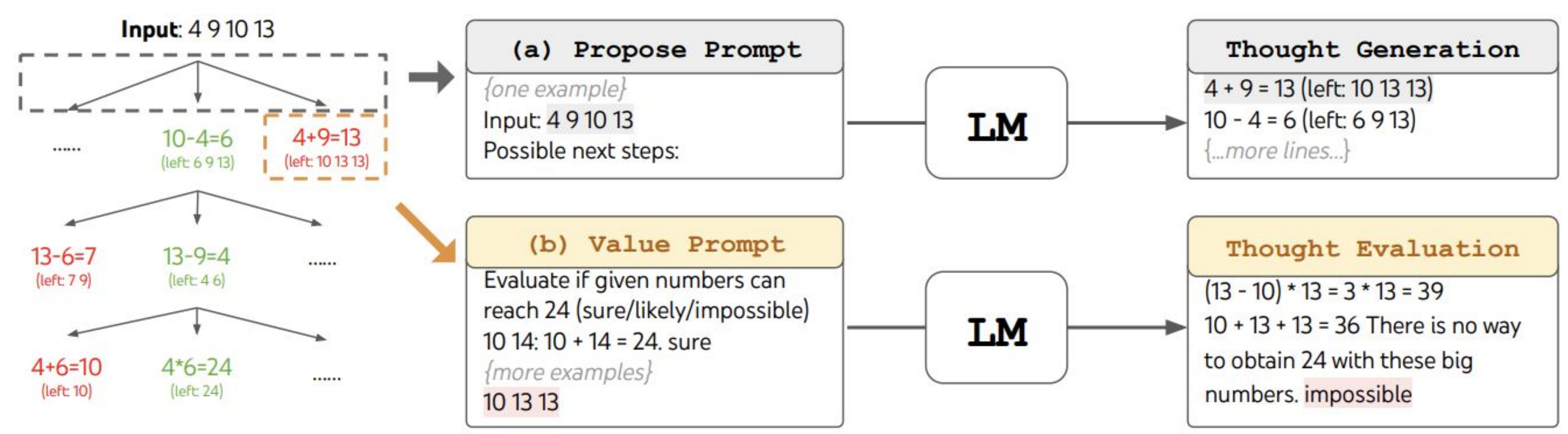
Table 3: Mini Crosswords results.

Methodology

1. Thought Decomposition
2. Thought Generation
3. State Evaluation
4. Search Algorithm

✦ Gemini 2.5 Flash

	Game of 24	Creative Writing	5x5 Crosswords
Input	4 numbers (4 9 10 13)	4 random sentences	10 clues (h1. presented;..)
Output	An equation to reach 24 (13-9)*(10-4)=24	A passage of 4 paragraphs ending in the 4 sentences	5x5 letters: SHOWN; WIRRA; AVAIL; ...
Thoughts	3 intermediate equations (13-9=4 (left 4,4,10); 10-4=6 (left 4,6); 4*6=24)	A short writing plan (1. Introduce a book that connects...)	Words to fill in for clues: (h1. shown; v5. naled; ...)
#ToT steps	3	1	5-10 (variable)



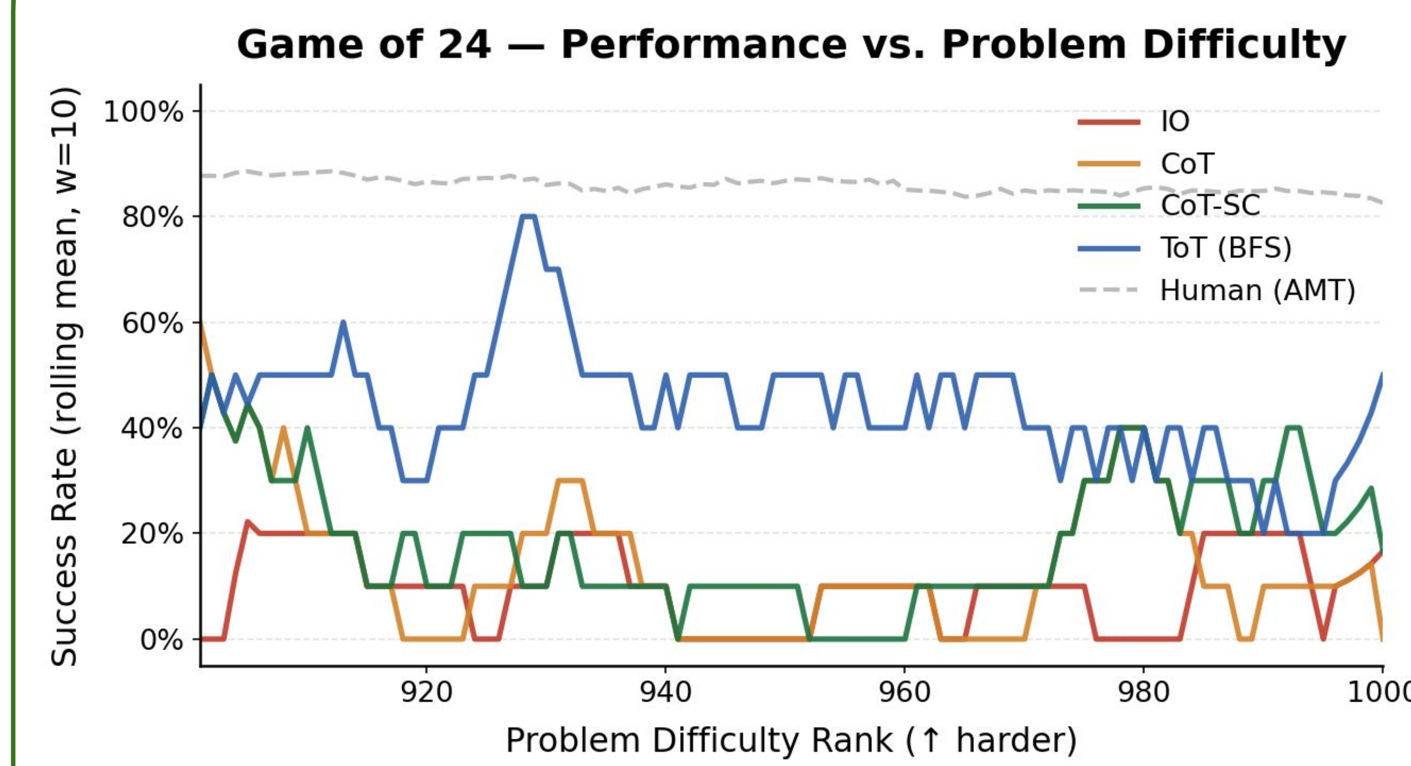
Conclusion / Future Work

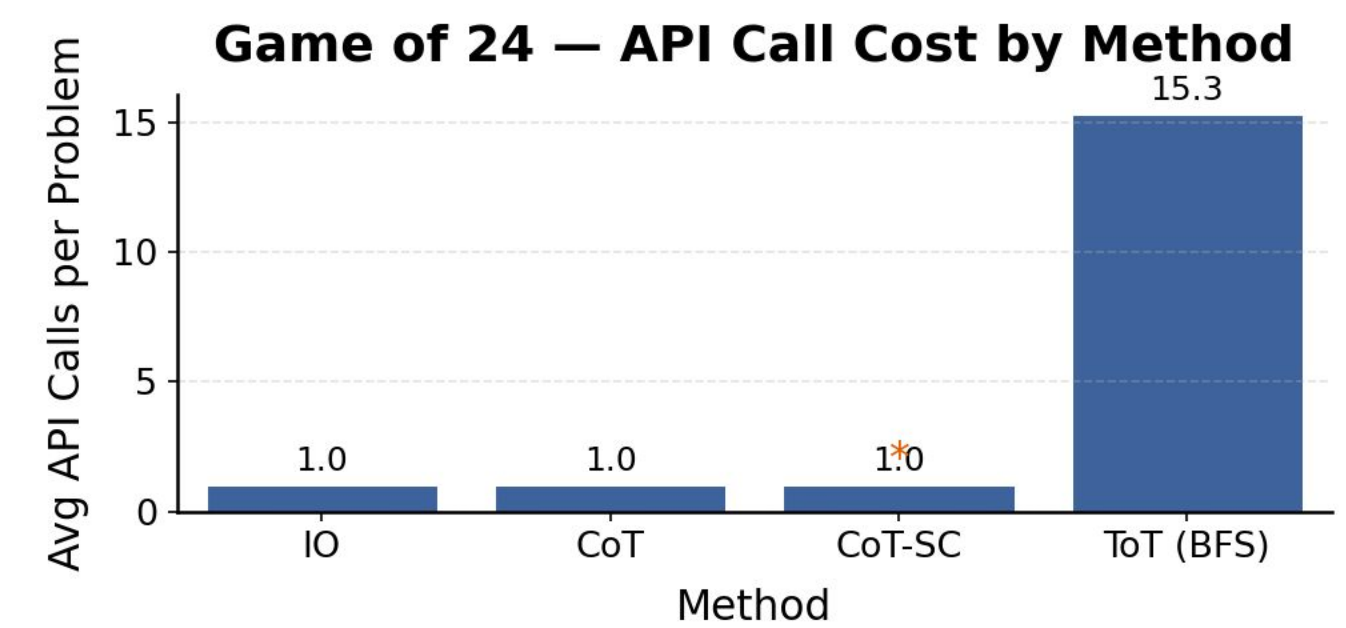
- ToT **significantly** improves success rate in tasks that require multi-step reasoning
- ToT implementation **significantly** increases costly resource usage (i.e. LLM API calls)
- ToT reasoning is **weakly correlated** with human accuracy (+0.168, p = 0.094)
- Model reasoning is **brittle**, even with ToT and 3 "thoughts"

Future Work:

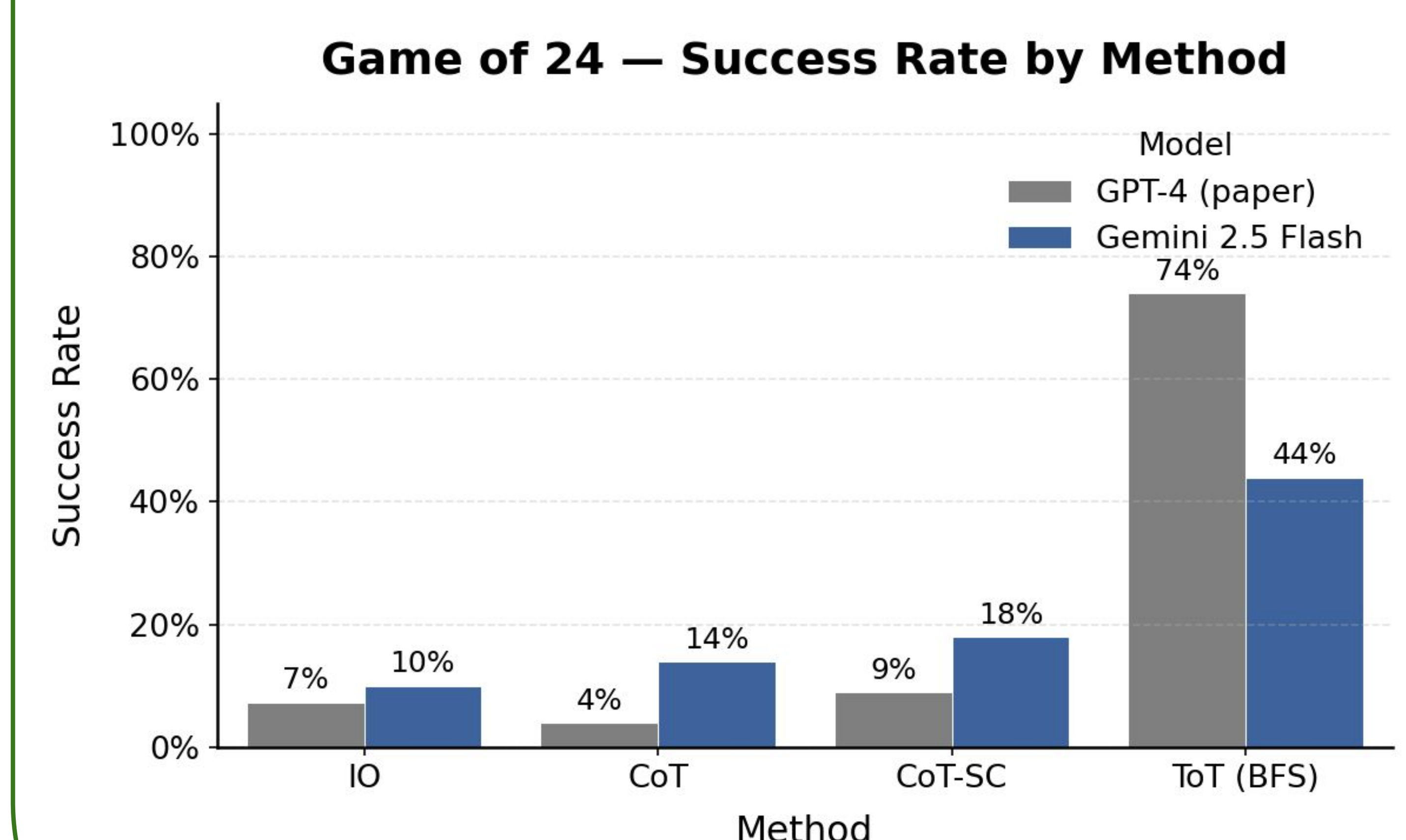
- Tuning the optimal number of search branches for a problem
- Utilizing mixed model types across a single problem to reduce resource requirements
- Analyze failure cases to determine cause of failure (e.g. arithmetic, heuristic errors)

Results

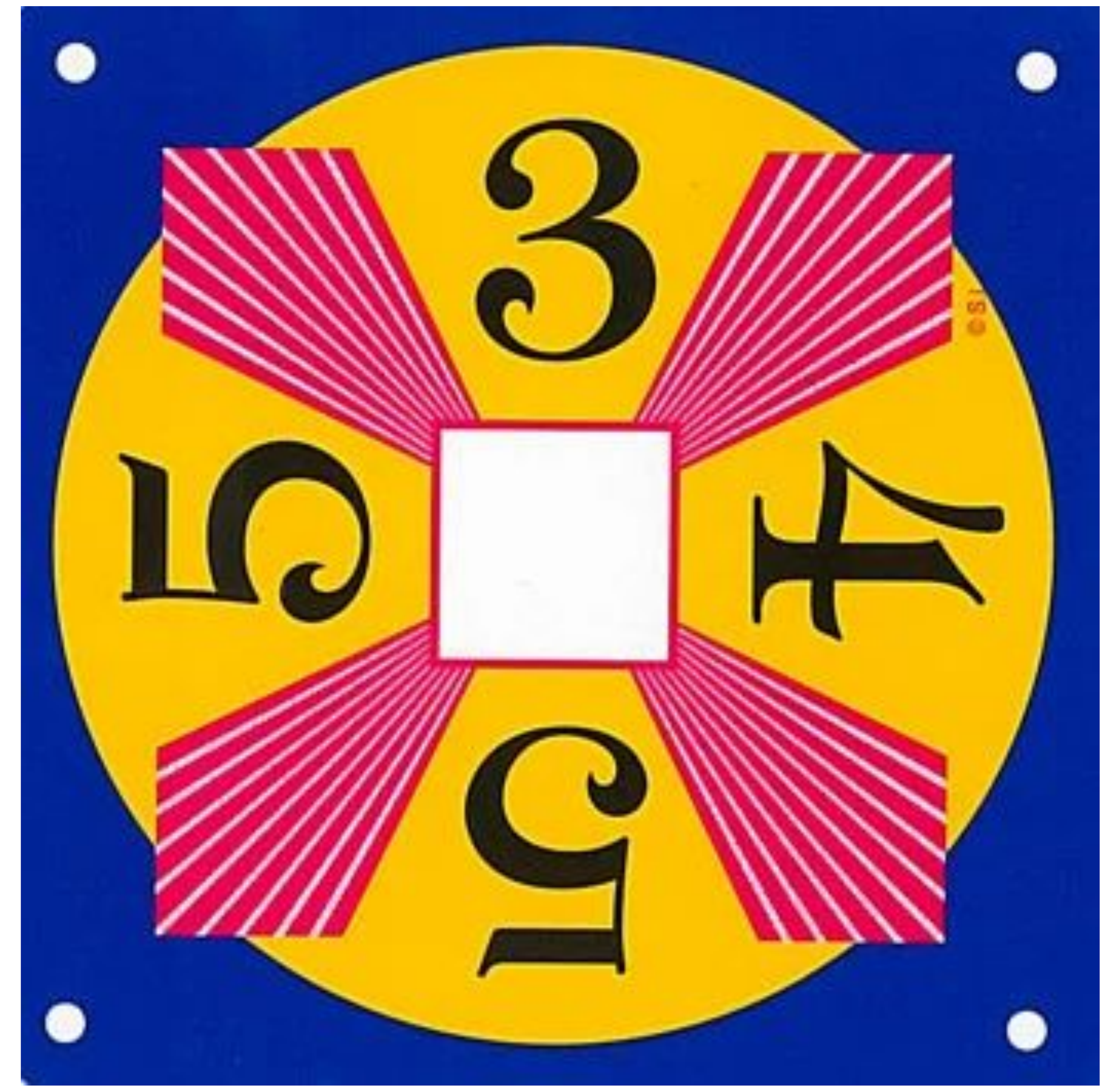




* CoT-SC true cost ≈ 5 calls/problem (Gemini batch counted as 1)



References: arXiv 2305.10601, https://github.com/princeton-nlp/tree-of-thought-llm, https://cameronrwolfe.substack.com/p/tree-of-thoughts-prompting



3. Poster Structure

Your poster should include the following sections:

1. **Title and Authors**
   - Title of your project
   - Names of all group members and institutional affiliation (You may utilize the headers and/or footer of your poster for this information to ensure optimal poster-space usage)
2. **Introduction / Background / Motivation**
   - Brief description of the problem being addressed.
   - Clear statement of the goal or hypothesis of your project.
   - Specific result you aimed to reproduce or investigate.
   - Context and motivation for your work.
   - Summary of the chosen paper and main contributions (if applicable).
3. **Methodology**
   - Overview of your approach, including models, datasets, and tools used.
   - Any modifications or design choices made.
   - Any interesting experiments your group came up with to add to the paper
4. **Results**
   - Present results in visual form: graphs, tables, charts, images.
   - Compare your outcomes with the original findings or expectations.
5. **Conclusion**

   - Summarize key takeaways from your work.
6. **Future Work (Optional)**
   - Mention possible directions for future exploration.
7. **References**
   - Cite the original paper and any resources used.